



Changing impact of fiscal policy on selected ASEAN countries[☆]

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ABSTRACT

This paper investigates the effectiveness of fiscal policy in five Association of Southeast Asian Nations: Indonesia, Malaysia, the Philippines, Singapore and Thailand. Through a small open economy structural vector autoregression model, government spending is found to have weak and largely insignificant impact on output, while taxes are found to have outcomes contrary to conventional theory. Extensions using a time-varying VAR model reveal that the positive impact from higher taxes on output mainly reflects heightened concerns over public finances during the Asian financial crisis and the recent global financial crisis. On the other hand, for Thailand, there is some evidence that government spending can at times be useful as a tool for short-term countercyclical policy.

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1. Introduction

The flurry of fiscal stimulus packages implemented in the last few years both in the developed countries and developing countries in response to the sharp economic slowdown seem to suggest an unequivocal support for fiscal policy in stabilizing economic fluctuations. The reality is quite the opposite. A debate hosted by *The Economist* amid the global recession in early 2009 on whether “Keynesian principles” should dominate policymaking of the day ended with a vote of 63% against the motion.² Meanwhile, despite the massive stimulus efforts, the pace of economic recovery especially in the advanced countries has been lackluster. In developing Asia, there has largely been an absence of the debate on the efficacy of fiscal stimulus. The fact that the region has recovered much faster than the rest of the world seems to suggest that the stimulative policies implemented have worked well. Yet surprisingly there is very little empirical evidence to conclusively say whether this was the case.³

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² See <http://www.economist.com/debate/overview/140>.

³ See ADB (2010, Part 2) for some recent work.

This paper asks the question whether, since the 1990s, fiscal policy in the five main Association of Southeast Asian Nations (ASEAN) (Indonesia, Malaysia, the Philippines, Singapore and Thailand) has been effective as a macroeconomic stabilizing tool. As part of this exercise, the paper also estimates the size of each economy's fiscal multiplier – how much a dollar spent by the government translates into a change in output. The paper extends the structural vector autoregression (SVAR) model by Blanchard and Perotti (2002) to capture the small open economy feature of the five selected economies. Blanchard and Perotti's model has been widely employed in the study of countercyclical fiscal policy for developed countries, but application to developing countries, in particular developing Asian economies, has been relatively scarce. However, a notable drawback of the approach is the inability of the model to examine a specific episode of countercyclical policy. Put differently, it only portrays an average estimate over the selected sample period. To do this, a time-varying VAR is employed to investigate possible changes to the effectiveness of fiscal policy over time. This is perhaps the first paper that applies a time-varying VAR model to study the evolution of fiscal policy for developing countries.

To preview the results, government spending is found to have no immediate and statistically significant impact on output in all the countries studied here. Yet for Singapore and Thailand, there is some evidence to show a stronger impact during the Asian financial crisis and the current global financial crisis. For Indonesia too, there has been an improvement in the effectiveness of government spending since the Asian financial crisis. On other hand, somewhat surprisingly, tax increases generate positive output growth in all the countries (a phenomenon commonly known as expansionary fiscal contraction in the literature), but this is only statistically significant in Thailand and Indonesia. In general, there seems to be greater evidence of the phenomenon during the Asian financial crisis and the global financial crisis, particularly in the Philippines and to a lesser extent Malaysia. For Indonesia, however, the puzzle seems to have waned in the 2000s, after the Asian financial crisis.

The rest of the paper is structured as follows. Section 2 reviews the theoretical and empirical literature on the study of the effectiveness of fiscal policy. Section 3 introduces the SVAR and time-varying VAR methodologies, while Section 4 delves into data and estimation issues. Section 5 presents and discusses the results, and Section 6 concludes.

2. Literature on the impact of fiscal policy

2.1. Theory

The effectiveness of fiscal policy as a tool of macroeconomic management has been widely debated, as many empirical studies find different and sometimes conflicting results. The theoretical literature itself prescribes different fiscal multiplier sizes. The Keynesian school generally argues that the spending multiplier is greater than one, while the neoclassical school says it is less than one. Table 1 summarizes the main theoretical results.⁴

The standard Keynesian multiplier of greater than one does not always hold, and may vary depending on, among other factors, the degree of monetary policy accommodation, the exchange rate regime, an economy's trade openness, and the extent of financial development. For example, it may be less than one in a flexible exchange rate regime, where higher interest rates caused by higher government spending leads to an appreciation of the domestic currency, which increases imports at the expense of exports. Similarly, if an economy is very open, the leakage of government spending via higher imports will be larger, and leads to a smaller multiplier. In a Ricardian equivalence world, fiscal spending is said to have no impact on output (Barro, 1974). Households who are forward-looking know that debt-financed government spending (or tax cuts) today translates into higher taxes in the future to pay for the higher debt, and hence the net effect on current output is zero.

There is another literature that posits the multiplier can in fact be negative, what is known as the non-Keynesian or expansionary fiscal contraction (or conversely, contractionary fiscal expansion) effect.⁵ Theories that explain this phenomenon appear to have followed the empirical work by, among others, Giavazzi and Pagano (1990), Alesina and Ardagna (1998, 2009), and Alesina and Perotti (1997). They relate to several core ideas such as credibility of fiscal policy, uncertainty, debt sustainability, and risk premium over government bonds. Blanchard (1990) formulates a simple theoretical argument to show that the benefit of early fiscal consolidation could increase households' total wealth by reducing the uncertainty of more costly and painful adjustment later on.⁶ Alesina and Ardagna (2009) add a further channel based on agents' expectations on changes in interest rate or risk premium on bonds. They show that a credible commitment to avoid a debt default or build-up in debt can lower agents' expectations of interest rate levels and the risk premium on government bonds. Hemming, Kell, and Mahfouz (2002) argue that if a fiscal expansion is linked to increased uncertainty and the government's credibility is questionable, precautionary behavior by households and firms will dominate and this contributes to the negative multiplier.

2.2. Empirics

Empirical studies on the effectiveness of fiscal policy are largely done via structural macroeconomic or VAR models. There is a wide variation of fiscal multiplier estimates – from less than zero to over four – depending on the type of identifying

⁴ For a more detailed discussion, please refer to the working paper version of this paper (<http://ideas.repec.org/p/ris/adbrei/0070.html>).

⁵ Occasionally, the term “anti-Keynesian” is also used, for example, Miller et al. (1990).

⁶ Blanchard provides a caveat that the effect is most likely in the face of high debt levels.

assumptions, the type of fiscal instrument, the country under study, and time periods (Spilimbergo, Symansky, Blanchard, & Cottarelli, 2008).

In studies using structural macroeconomic models, the short-run multiplier is often positive, the spending multiplier is larger than the tax multiplier, and both are larger when fiscal expansion is accommodated by monetary policy. However, in the long-term, the multiplier can be negative because of crowding out effects. Hemming et al. (2002) summarize the results based on some well-known models such as the IMF Multimod model, the OECD Interlink model and the McKibbin-Sachs (MSG) model involving the US, Germany and Japan.⁷ They find that the short-term multiplier ranges from 0.6 to 1.4 for the spending multiplier, and 0.3 to 0.8 for the tax multiplier. More recently, using the IMF Global Integrated Monetary and Fiscal Model (GIMF), Freedman, Kumhoff, Laxton, and Lee (2009) find that when all regions undertake fiscal stimulus together – global policy coordination – the world spending multiplier is 1.7 and tax multiplier is 0.3.⁸ For example, for emerging Asia, if only the region undertakes its own fiscal expansion, the spending multiplier declines to 1.1 from 2.9. That said, in the long-run, if there is a permanent increase in the ratio of debt-to-GDP in the world as a result of the fiscal expansion, there will be a permanent 1.3% contraction in GDP.

Findings from the VAR based approach are generally positive but less than one for the spending multiplier and an even lower value for the tax multiplier. For developing countries, the multipliers are even lower. Blanchard and Perotti's (2002) SVAR for the US shows an impact spending multiplier of 0.9 and a tax multiplier of 0.7. Mountford and Uhlig's (2009) sign-restriction VAR produces lower values of 0.6 and 0.3, respectively. Perotti (2004) analyzes five OECD countries dividing them into two periods and find that only the US has a spending multiplier larger than one in the pre-1980 period. That said, the multiplier has declined over time for all countries in the sample. Ilzetzi, Mendoza, and Vegh (2009) look at 45 countries (20 advanced and 25 developing) and find the impact spending multipliers from different experiments to be consistent with theories: advanced countries, 0.24 versus developing countries, 0.04; fixed exchange rate regime, 0.2 versus flexible exchange rate regime, −0.04; closed economies, 0.26 versus open economies, −0.05. In addition, the more indebted developing countries (defined as debt-to-GDP ratio above 50%) have a spending multiplier that turns negative after only four quarters.

There are only a few studies that examine Asia. Eskesen (2009) and Jha et al. (2010) are closest to our study from a methodological perspective.⁹ Eskesen only examines Singapore using the Blanchard and Perotti SVAR, while Jha et al. use the Mountford and Uhlig sign restrictions VAR on a larger sample of 10 developing Asian countries including those in our sample. They find a largely weak and insignificant impact of government spending on output. Jha et al. also find an expansionary fiscal contraction (in terms of a positive tax shock) on output in the People's Republic of China; Singapore; Taipei, China; Thailand. In Eskesen's paper, although the response of output to a tax shock presented shows a rise, the author makes no reference to whether this relates to an expansionary fiscal contraction or that a negative tax shock has been invoked.¹⁰

3. Methodology

3.1. Blanchard and Perotti SVAR

A distinguishing feature of the Blanchard and Perotti SVAR framework is that it relies on institutional information on the tax and transfer system and the timing of tax collections to separate the automatic response of taxes and spending to economic activity, and in doing so, isolate the effects of discretionary fiscal shocks. There are two particular strengths to this approach. First, discretionary fiscal policy is generally exogenous to output. Unlike monetary policy, fiscal variables are affected by many factors of which output stabilization is seldom the main driver. Second, at high enough frequency and because of the long lags of fiscal policy implementation, there is very little or no discretionary response of fiscal policy to unexpected changes in economic activity within the quarter.

The identification restrictions of the Blanchard and Perotti SVAR can be expressed as a class of AB SVAR model in Amisano and Giannini (1997):

$$A\varepsilon_t = BU_t, \quad \begin{pmatrix} 1 & 0 & a_{13} \\ 0 & 1 & a_{23} \\ a_{31} & a_{32} & 1 \end{pmatrix} \begin{pmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \\ \varepsilon_{3t} \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} & 0 \\ b_{21} & b_{22} & 0 \\ 0 & 0 & b_{33} \end{pmatrix} \begin{pmatrix} u_{1t} \\ u_{2t} \\ u_{3t} \end{pmatrix}. \quad (1)$$

⁷ See also Coenen et al. (2010) who examine a variety of more recent national and global macro models on the US and the euro area/European Union and draw similar conclusions.

⁸ Only the first year multiplier is presented. The model comprises five countries/regions: the US, the euro area, Japan, emerging Asia and the rest of the world. The authors simulate the model with two years of fiscal stimulus based on four different fiscal instruments: lump sum transfers; labor income tax; government investment; and targeted transfers. With monetary accommodation, the targeted transfers multiplier is 1.1, and the lump sum transfers, 0.5.

⁹ We also find two other studies which use a different methodology. Chang et al. (2002) use a cointegration approach and find that for the three countries they study (Thailand; Republic of Korea; and Taipei, China), fiscal policy has no impact on output growth. Ducanes et al. (2006) apply a structural model on four countries (Bangladesh, People's Republic of China, Indonesia and the Philippines). They find that government spending is more useful than tax cuts, and it is more effective in the Philippines than in Indonesia, but even so, all multipliers are less than one.

¹⁰ Eskesen's results do not include confidence band.

A is a $n \times n$ matrix of contemporaneous relations among the variables; ε_t is the vector of the normally independently distributed reduced form residuals with variance–covariance $E(\varepsilon_t \varepsilon_t') = \Sigma$; B is a $n \times n$ matrix that allows some shocks to affect more than one endogenous variable in the model; U_t is the vector of structural shocks of policy and non-policy variables, where $U_t \sim N(0, I_n)$ and $E(U_t U_s) = 0$ for $t \neq s$.

Expanding the above matrix and denoting the three variables as taxes (tx), government spending (s), and real GDP (y) gives:

$$\varepsilon_t^{tx} = a_{13}\varepsilon_t^y + b_{11}u_t^{tx} + b_{12}u_t^s, \quad (2)$$

$$\varepsilon_t^s = a_{23}\varepsilon_t^y + b_{21}u_t^{tx} + b_{22}u_t^s, \quad (3)$$

$$\varepsilon_t^y = a_{31}\varepsilon_t^{tx} + a_{32}\varepsilon_t^s + b_{33}u_t^y. \quad (4)$$

Eqs. (2)–(4) denote the relationships among the reduced form residuals (unexpected movements) in taxes (ε_t^{tx}), spending (ε_t^s) and output (ε_t^y), as well as fiscal shocks (u_t^{tx} and u_t^s) and output shock (u_t^y). Eqs. (2) and (3) say tax and spending activities are dependent on economic activity and shocks to fiscal policy (u_t^{tx} and u_t^s), as to be expected. Eq. (4) says economic activity is itself influenced by tax and spending activities as well as shocks to itself.

From matrices A and B , there are nine parameters that need to be estimated based on six knowns (given Σ has $n(n+1)/2$ unique elements), which means a further three restrictions have to be imposed. First, Blanchard and Perotti estimate a_{13} , the elasticity of taxes to GDP, separately. They follow the calculations of [Giorno, Richardson, Roseveare, and van der Noord \(1995\)](#), which depend on the summation of the elasticity of each tax category weighted by the tax base. Here, due to the lack of disaggregated information on different tax categories, a_{13} is estimated from a simple regression of taxes on GDP. Second, they assume a_{23} , the elasticity of government spending to GDP, is zero. This essentially assumes there is no automatic feedback from economic activity to government purchases of goods and services within the quarter. It is important to recognize that both a_{13} and a_{23} can potentially be affected by the discretionary fiscal policy changes with respect to economic shocks. Yet realistically due to the lag in policy implementation, it is unlikely either a_{13} or a_{23} can capture the discretionary fiscal policy effects especially with the use of quarterly data.¹¹ Third, they decide to let b_{12} be zero, implying that tax decisions are made before spending decisions. This means a tax shock affects spending contemporaneously and not the reverse. Since there is no straightforward way to determine this, they also try the reverse, that is, by letting b_{21} be zero so that b_{12} can be estimated. They find the overall results do not change significantly either way, and we reach the same conclusion.¹² Hence, for simplicity, we continue to let b_{12} be zero.

To capture the small open economy feature, see for example, [Dungey and Fry \(2009\)](#), we extend the baseline Blanchard and Perotti SVAR by including world GDP as a contemporaneous exogenous variable. The inclusion of the exogenous variable does not affect the Blanchard and Perotti identification strategy. For comparison, we also present the results based on the Cholesky ordering. In the Cholesky VAR, the endogenous variables are positioned in the same order as in the Blanchard and Perotti SVAR: tx , s and y . If A is a lower triangular matrix with the value of one at the diagonal elements and B , a diagonal matrix, then the Cholesky matrix, $C = A^{-1}B$, is the lower triangular matrix with the diagonal elements being the standard deviation of each structural shock.

3.2. Time-varying VAR model

To investigate the time-varying effects of fiscal policy over the sample, the VAR model outlined above is extended in two important dimensions. First, we allow for time variation in the autoregressive coefficients of the VAR model. Second, we allow for stochastic volatility in the reduced-form error terms. The reduced-form time-varying VAR can be summarized as:

$$Y_t = B_{1,t}Y_{t-1} + \dots + B_{p,t}Y_{t-p} + CZ_t + \varepsilon_t \equiv X_t'\theta_t + \varepsilon_t \quad (5)$$

Following [Cogley and Sargent \(2005\)](#) and [Primiceri \(2005\)](#), the VARs time-varying parameters, collected in the vector θ_t , are postulated to evolve as a random walk such that:

$$\theta_t = \theta_{t-1} + \eta_t \quad (6)$$

with $\eta_t \sim N(0, Q)$. Note world GDP is treated as an exogenous variable (Z_t), where the coefficient has the same time-varying property as the endogenous variables. The VARs reduced-form innovations in (5) are postulated to be zero-mean and normally distributed, with time-varying covariance matrix Ω_t , which can be factored as:

$$\text{Var}(\varepsilon_t) \equiv \Omega_t = A_t^{-1}H_t(A_t^{-1})' \quad (7)$$

¹¹ Theoretically, a_{13} and a_{23} can capture both the automatic effects of economic activity on taxes and government spending as well as the discretionary fiscal response to unexpected economic events. Nevertheless, the use of quarterly data effectively implies only the automatic effects of economic activity are accounted for. And since a_{23} is assumed to be zero in the model, a_{13} essentially captures only the automatic effects of economic activity on taxes.

¹² These results are available upon request.

The time-varying matrices H_t and A_t are defined as:

$$H_t = \begin{pmatrix} h_{1,t} & 0 & 0 \\ 0 & h_{2,t} & 0 \\ 0 & 0 & h_{3,t} \end{pmatrix} \quad \text{and} \quad A_t = \begin{pmatrix} 1 & 0 & 0 \\ a_{21,t} & 1 & 0 \\ a_{31,t} & a_{32,t} & 1 \end{pmatrix}, \quad (8)$$

with $h_{i,t}$ evolving as geometric random walks:

$$\ln h_{i,t} = \ln h_{i,t-1} + \nu_{i,t} \quad (9)$$

Following Primiceri (2005), we postulate the non-zero and non-one elements of the matrix A_t (collected in the vector $\alpha_t \equiv [\alpha_{21,t}, \alpha_{31,t}, \alpha_{32,t}]'$) to evolve as driftless random walks:

$$\alpha_t = \alpha_{t-1} + \tau_t \quad (10)$$

Finally, we assume the vector of error terms $[u'_t, \eta'_t, \tau'_t, \nu'_t]'$ of the system to be distributed as

$$\begin{pmatrix} u_t \\ \eta_t \\ \tau_t \\ \nu_t \end{pmatrix} \sim N(0, V), \quad \text{with } V = \begin{pmatrix} I_3 & 0 & 0 & 0 \\ 0 & Q & 0 & 0 \\ 0 & 0 & S & 0 \\ 0 & 0 & 0 & Z \end{pmatrix} \quad \text{and} \quad Z = \begin{pmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{pmatrix} \quad (11)$$

where u_t is the structural error terms of the VAR such that $\varepsilon_t = A_t^{-1} H_t^{1/2} u_t$. The block-diagonal structure of V assumes that the idiosyncratic error terms of the system are uncorrelated with each other. Lastly, we also adopt the additional simplifying assumption of postulating a block-diagonal structure for S , namely:

$$S \equiv \text{Var}(\tau_t) = \begin{pmatrix} S_1 & 0_{1 \times 2} \\ 0_{2 \times 1} & S_2 \end{pmatrix} \quad (12)$$

with $S_1 \equiv \text{Var}(\tau_{21,t})$, and $S_2 \equiv \text{Var}([\tau_{21,t}, \tau_{32,t}]')$. This implies that the non-zero and non-one elements of A_t belonging to different rows (equations of the VAR) evolve independently. As discussed in Primiceri (2005), this assumption drastically simplifies inference and allows the estimation procedure to apply Gibbs sampling on the non-zero and non-one elements of A_t equation by equation. We apply the same identification assumptions as in the time-invariant VAR described earlier.

We estimate Eqs. (5)–(12) using Bayesian methods assuming the priors in Primiceri (2005). The basic steps of the algorithm are as follows.¹³ First, given the initial values for A_t and H_t , we simulate the VAR parameters and hyperparameters. Next, conditional on the VAR parameters, stochastic volatilities ($h_{i,t}$) are drawn. Finally, conditional on all other parameters, we draw the elements of A_t equation by equation. The algorithm uses 250,000 Gibbs sampling replications and discards the first 200,000 as burn-in. We retain 1,000 draws after the burn-in sample by including every 50th draw for the computation of the impulse response functions described in Koop, Pesaran, and Potter (1996). The posterior moments appear stationary over the retained draws providing evidence of convergence. Given the estimated parameters of the model, we simulate the time-varying IRF's using the Blanchard and Perotti restrictions described in Section 3.1.

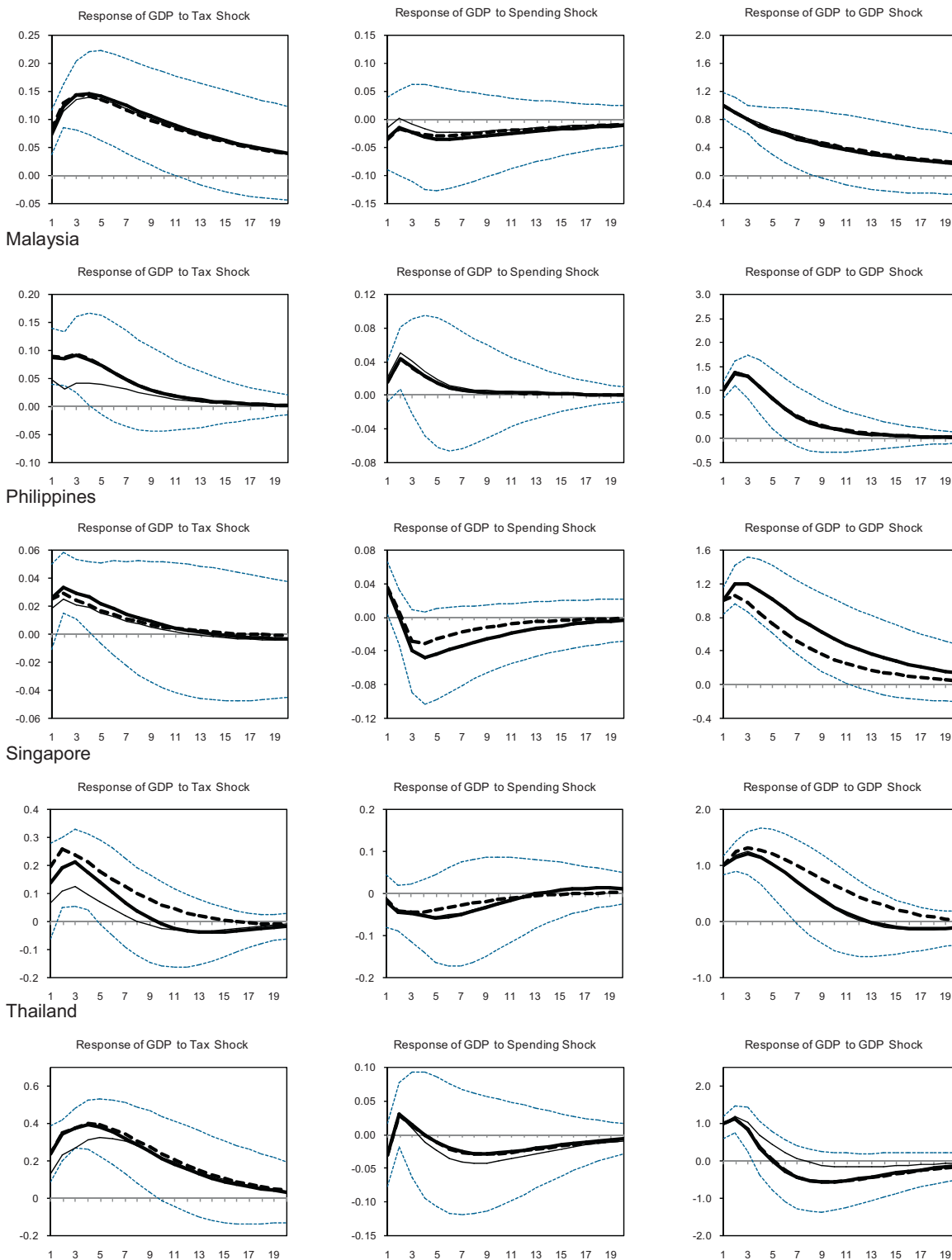
4. Data and estimation issues

Quarterly data from mostly 1990:1 to 2009:4 for the five main ASEAN countries (Indonesia, Malaysia, the Philippines, Singapore and Thailand) are collected from various sources such as the CEIC database, the World Bank's *World Development Indicators* and websites of national statistical offices/central banks. For detailed data sources and descriptions, please see the appendix. Blanchard and Perotti (2002) define net taxes as total tax revenue minus grants and interest payments. Nonetheless, due to data limitations and the fact we try to use the longest available series, none of our tax variable meets this definition. Therefore, for Indonesia, the tax variable refers to total government revenue; Malaysia and the Philippines, total tax revenue minus interest payments; Singapore and Thailand, total tax revenue. For GDP and government spending, the data are obtained from the respective national accounts. Due to the unavailability of the breakdown on government investment, government consumption is taken to mean government spending.¹⁴

Following Blanchard and Perotti (2002), all data are expressed in natural logarithm, real and per capita terms. We seasonally adjust all the data in their natural log form using the standard X11 methodology. Then we express them in per capita terms. The annual population data from *World Development Indicators* are converted to quarterly data using a simple linear interpolation. Since the population data end in 2008, the 2009 projection is based on the average growth rate from 2003 to 2008. The GDP deflator is obtained from taking the ratio of nominal and real GDP are subsequently used to deflate the data. Lastly, the data are linearly detrended – hence, variations in the transformed data are in essence movements around a long-run equilibrium.

¹³ See Kim and Nelson (1999) for more details.

¹⁴ Quarterly investment data from national accounts are only available for Thailand. In the spirit of Blanchard and Perotti (2002), we did not use current and development expenditure data from fiscal accounts. Jha et al. (2010) use the latter and produce largely similar results to ours especially in the context of government spending shock on output.



Note: The size of shock is one unit or one percentage point. The thick line refers to the impulse response of the Blanchard and Perotti SVAR with estimated elasticity of taxes to GDP. The thin line is the average of the impulse responses of the Cholesky VAR and the Blanchard and Perotti SVAR with the elasticity of taxes to GDP of 0.5. The thick dashed line refers to the impulse response of the small-open economy Blanchard and Perotti SVAR. The spaced lines refer to the $\pm$ two asymptotic standard-deviation error bands of the standard Blanchard and Perotti SVAR. Source: Authors' calculations.

Fig. 1. Impulse response functions of GDP to all shocks. Note: The size of shock is one unit or one percentage point. The thick line refers to the impulse response of the Blanchard and Perotti SVAR with estimated elasticity of taxes to GDP. The thin line is the average of the impulse responses of the Cholesky

Table 1
Theoretical predictions of fiscal (spending) multiplier.

Scenario	Size of multiplier
Keynesian	
1. Standard Keynesian spending multiplier	>1
(Tax multiplier) ^a	<1 or >1
Simple Keynesian extensions	
2. Accommodative monetary policy	>1
3. Liquidity trap (interest rate at the zero bound)	>1
4. Flexible exchange rates	<1
5. Pegged exchange rates	>1
6. Highly open economy	<1
Neoclassical	
7. Neoclassical model	<1
Other considerations	
8. Complete Ricardian equivalence	0
9. Debt/fiscal sustainability issues	<1
10. Expansionary fiscal contractions	<0

Sources: Baxter and King (1993), Barro (1974), Mankiw (2000), Hemming et al. (2002), IMF (2008), Alesina and Ardagna (2009), and Woodford (2010).

^a The standard Keynesian spending multiplier is equal to $1/(1 - MPC)$, where $0 < MPC$ (marginal propensity to consume) < 1 . Hence, the spending multiplier is ≥ 1 . On the other hand, the tax multiplier $= -MPC/(1 - MPC)$. Hence, it is larger than 1 if $MPC > 0.5$, and smaller if $MPC < 0.5$.

In the Cholesky and SVAR estimations, the same lag length is used based on the most commonly recommended lag order by the standard information criteria of sequential modified likelihood ratio, Final Prediction Error, Akaike, Schwarz and Hannan-Quinn. For Indonesia, the lag order is one; Malaysia, the Philippines and Thailand, two; Singapore, three. Both the Cholesky and SVAR are also estimated with a constant. The same lag lengths are used for the time-varying VAR models. Model stability tests show the model is stable for all countries – all roots have modulus less than one. Furthermore, the LM test does not indicate significant autocorrelations in the estimated residuals.

The estimated elasticity of taxes to GDP, based on a simple regression between the two variables, for Indonesia is 0.63; Malaysia, 1.61; Philippines, 1.04; Singapore, 1.10; Thailand, 1.90. As a robustness check, we also compare the results when the elasticity is 0.5, a number Eskesen (2009) uses for Singapore.

5. Results

5.1. Basic model

Fig. 1 shows each country's impulse response function of GDP to the three shocks of taxes, government spending and GDP, respectively. Each shock is normalized to have a contemporaneous impact of one-percentage point increase. As comparison, besides the results from the Blanchard–Perotti SVAR with estimated elasticity of taxes to GDP, results for the average impulse response of the same SVAR but with elasticity of net taxes to GDP (a_{13}) of 0.5 and the Cholesky VAR, and the small-open economy Blanchard and Perotti SVAR are also presented.

In general, the overall results are not sensitive to any particular specification. And whether the elasticity of net taxes to GDP is estimated or assumed to be 0.5 does not seem to matter a great deal. Furthermore, there is little difference between the standard SVAR and the open-economy version. Three key results are evident. First, the impulse response of GDP to a positive tax shock produces a rise in GDP in all economies. The effect seems particularly strong for Indonesia and Thailand, as they are both significant for ten quarters and peak at between two to four quarters across different model specifications. Second, in all countries, by and large, the impulse responses of GDP to a government spending shock are close to zero and statistically insignificant. Statistical insignificance aside, there are notable variations in the dynamics of the impulse response which will be discussed below. Third, the impact fiscal multiplier of either the tax shock or the government spending shock (summarized in Tables 2 and 3) is generally very small and less than one. The result is significant only in case of the tax shock, and only in Indonesia, Thailand and Malaysia (when the elasticity of net taxes is calculated). The only country with a multiplier greater than one, be it tax or spending, is that of Thailand with respect to the tax shock and when a_{13} is estimated.

It is unclear why we only find a consistent expansionary impact on output with the positive tax shock, but not with spending cut. One reason could be that fiscal adjustments have been more commonly carried out through tax changes. Makin

VAR and the Blanchard and Perotti SVAR with the elasticity of taxes to GDP of 0.5. The thick dashed line refers to the impulse response of the small-open economy Blanchard and Perotti SVAR. The spaced lines refer to the $\pm$ two asymptotic standard-deviation error bands of the standard Blanchard and Perotti SVAR.

Source: Authors' calculations.

Table 2

Impact fiscal multiplier from a tax shock.

	Cholesky VAR	Blanchard and Perotti SVAR with estimated α_{13}	Blanchard and Perotti SVAR with $\alpha_{13} = 0.5$
Indonesia	0.37 ^a	0.45 ^a	0.43 ^a
Malaysia	0.29	0.65 ^a	0.41
Philippines	0.17	0.27	0.22
Singapore	0.32	0.94	0.61
Thailand	0.59 ^a	1.22 ^a	0.78 ^a

Source: Authors' calculations.

Note: To calculate the impact fiscal multiplier, the impact response of GDP to the tax shock is divided by the ratio of average net tax revenue to average GDP over the sample period. For Indonesia, the ratio is 0.17; Malaysia, 0.14; the Philippines, 0.10; Singapore, 0.15; Thailand, 0.19.

^a Indicates the impact impulse response is statistically significant from zero at the 5% level.

Table 3

Impact fiscal multiplier from a spending shock.

	Cholesky VAR	Blanchard and Perotti SVAR with estimated α_{13}	Blanchard and Perotti SVAR with $\alpha_{13} = 0.5$
Indonesia	−0.01	−0.42	−0.34
Malaysia	0.19	0.14	0.17
Philippines	0.31	0.32	0.32
Singapore	−0.13	−0.18	−0.16
Thailand	−0.25	−0.26	−0.27

Source: Authors' calculations.

Note: To calculate the impact fiscal multiplier, the impact response of GDP to the spending shock is divided by the ratio of average government consumption to average GDP over the sample period. For Indonesia, the ratio is 0.08; Malaysia, 0.12; the Philippines, 0.11; Singapore, 0.10; Thailand, 0.11.

(2005) argues that there is less scope for most of these countries to curtail growth-enhancing or productivity-driven expenditures such as transport infrastructure, power, telecommunications and education. Being developing countries, these expenditures are essential to future economic development. In contrast, the revenue shares of these countries are much lower than international standards. Hence, there is ample room to broaden revenue bases through new tax policies and administrative reforms.

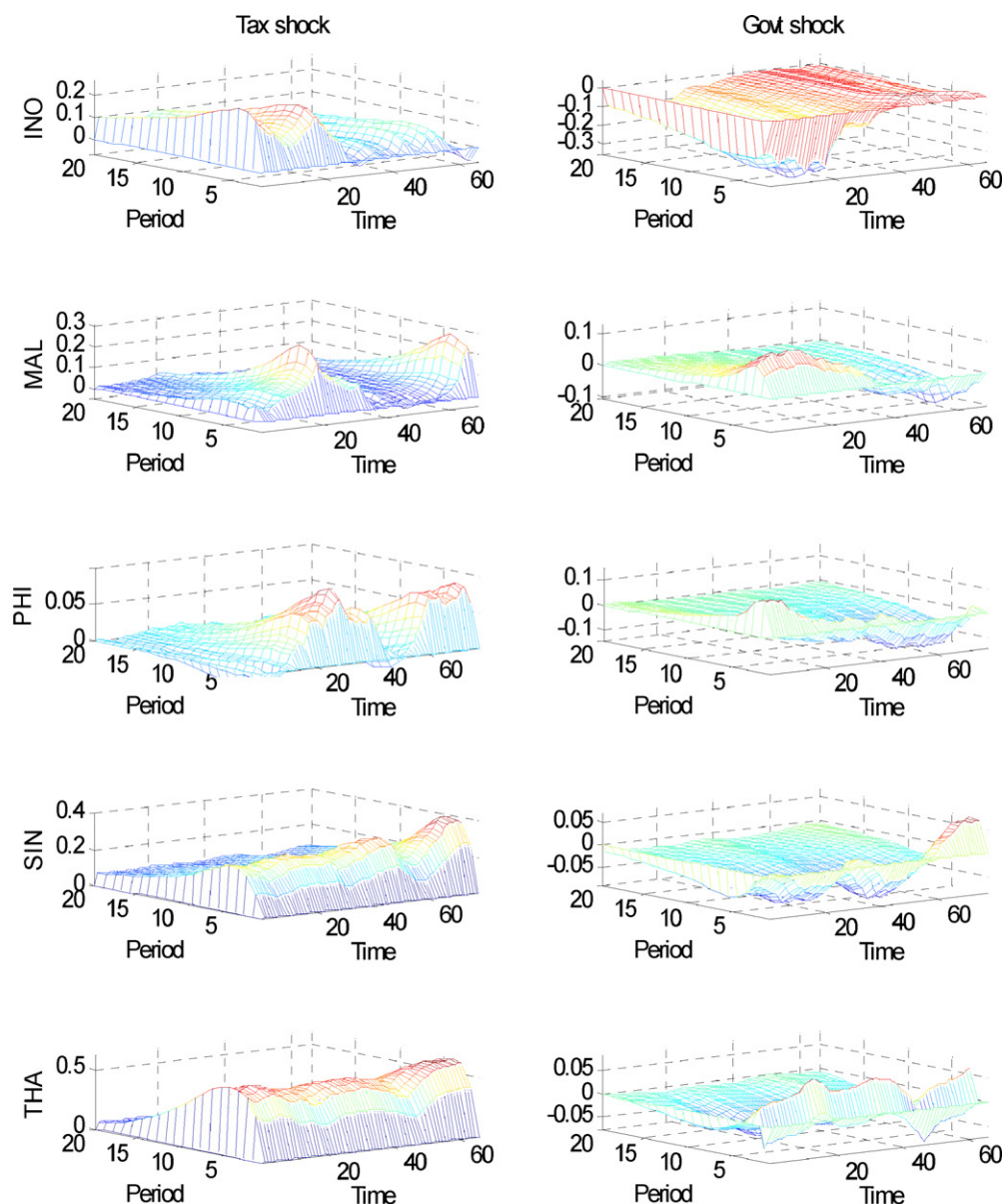
For small and open economies, the leakage of fiscal spending through higher imports is a key channel that negates the stimulative effects. This factor is likely to be most relevant for Singapore, Malaysia and Thailand, each being a small and highly open economy with a share of total trade to GDP of 283%, 146% and 108%, respectively, in 2009. Philippines's ratio is slightly lower at 51%, and Indonesia's, 39%. This is in line with the findings by Ilzetzki et al. (2009). In their study, an open economy is defined as having a share of total trade to GDP of over 60%. For this group, they find both the impact and the long-run responses of output to a government spending shock are not statistically different from zero.

The choice of exchange rate regime may have also weakened the multiplier effects. Apart from Malaysia which fixed its exchange rate for a brief period, most other countries have adopted a more flexible exchange rate regime particularly since the Asian financial crisis.¹⁵ The finding by Ilzetzki et al. (2009) lends support to this view, where the output response to a government spending shock of economies adopting a flexible exchange rate regime are found to be weakly negative and statistically insignificant.

Generally, there might have been some degree of monetary policy accommodation, yet interest rates were never allowed to fall too much due to other concerns and they were never close to the zero bound. This lack of monetary policy accommodation could have mitigated or even reversed the expansionary fiscal impact. While it is true that policy interest rates in the ASEAN5 were reduced in the face of the current global financial crisis, they were, however, not as large as the reductions in the US, which have hit the zero bound since late 2008. Meanwhile, during the Asian financial crisis, interest rates were kept relatively high even up to the end of 1998 to allay fears of further capital outflows and to reinforce confidence in domestic asset holdings. This happened in spite of an already contracting output seen as early as the second quarter of 1997 in Thailand and in the first half of 1998 in the other countries. Interest rates did fall later on but the recovery from the crisis for these countries was largely driven by the quick turnaround in external demand, where domestic interest rates had little or no influence.

Persistent and large budget deficit to GDP balances increase debt levels and this brings the issue of debt sustainability to the forefront, especially during crisis times. Apart from Singapore, the budget balance-to-GDP ratios of the other countries have remained in deficit since the Asian financial crisis. While they may not be very large, they have persisted and have worsened during the global financial crisis. This is true even before the 1990s, again with the exception of Singapore and, to a lesser extent, Thailand. That said, Indonesia and Thailand, in contrast to Malaysia and the Philippines, had made greater progress in reducing the deficits in recent years, despite being the worst hit countries during the Asian financial crisis.

¹⁵ See for example ADB (2007, Box 2).



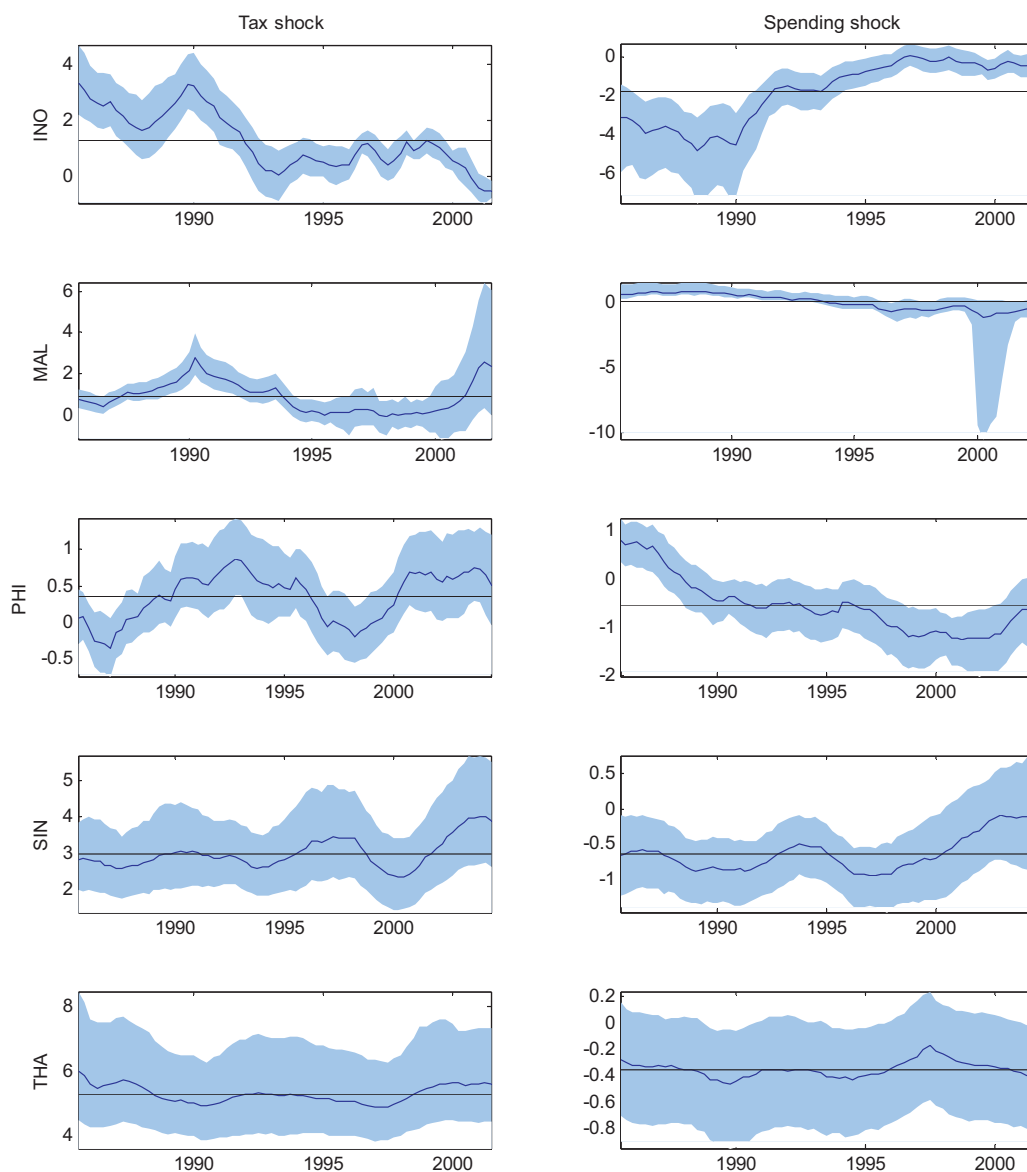
Note, the x-axis represents the date from 1990–2009, the y-axis corresponds to the IRF horizon, percentage points. Source: Authors' calculations.

Fig. 2. Time-varying impulse responses of GDP to tax and spending shocks. Note: The x-axis represents the date from 1990 to 2009, the y-axis corresponds to the IRF horizon, and the z-axis is the impact in percentage points. Source: Authors' calculations.

Considering even for the developed countries where fiscal situations can deteriorate markedly very quickly, the overall evidence does not provide overwhelming support that the countries have put paid to the issue of debt sustainability. This lends support to the expansionary fiscal contraction phenomenon.¹⁶

Unfortunately our structural VAR cannot capture the effects of pre-announced fiscal policy measures. One way to do this would be through a fully specified structural model, such as the IMF GIMF model, that incorporates agent's expectations.

¹⁶ For more detailed discussion, please refer to the working paper version.



Note: The horizontal line in each plot refers to the average cumulative response over the sample period. The error bands are the 95% confidence intervals. The kinked error bands for Malaysia around 2001 is due to draws being very close to unit root. Source: Authors' calculations.

Fig. 3. Two-year cumulative impulse response of GDP to tax and spending shocks. Note: The horizontal line in each plot refers to the average cumulative response over the sample period. The error bands are the 95% confidence intervals. The kinked error bands for Malaysia around 2001 are due to draws being very close to unit root.

Source: Authors' calculations.

Nevertheless, the structural shocks identified by our empirical model (as in [Giordano, Momigliano, Neri, & Perotti, 2007](#)) are consistent with the assessments documented in various ADB and IMF Article IV country reports.¹⁷

5.2. Time-varying VAR

[Fig. 2](#) shows the time-varying response of GDP to a positive one-unit tax and government spending shock across the five countries, respectively; [Fig. 3](#) presents the cumulative GDP response at the 2-year horizon to show the cumulative impact

¹⁷ To conserve space, we have not included the plots of structural shocks here but these are available upon request.

Table 4

Summary of overall results: impulse response of output.

	Indonesia	Malaysia	Philippines	Singapore	Thailand
Blanchard and Perotti SVAR (impact response)					
Tax shock	+	+	+	+	+ ^a
Spending shock	–	+	+	–	–
Time-varying VAR					
Tax shock	Positive in the 1990s, close to zero or somewhat negative afterward	Most prominent during the Asian and recent global crisis	Similar to Malaysia	Most prominent during current crisis	Positive and stable over the sample period
Spending shock	Mostly negative but has improved in the late 1990s	Positive prior to 2000 and negative afterwards	Positive before Asian crisis and negative afterwards	Mostly negative, somewhat better during the two crises.	Not much impact

Source: Authors' calculations.

^a Refers to being statistically significant.

more clearly. These together with earlier results are summarized in Table 4. Note, world GDP is treated as exogenous for the small open economy.

In Indonesia, the largely positive impact on output of a tax shock in the 1990s may have to do with the accommodative monetary policy stance taken by the central bank. As Nasution (2003) points out, “the central bank can assist fiscal sustainability by maintaining relatively low interest rates to encourage economic recovery and lighten the government’s interest burden without sacrificing monetary objectives” (p. 155). Our Cholesky VAR result incorporating an additional policy rate variable supports this as a positive tax shock coincides with a decline in the policy rate.¹⁸ Since 2000, however, the expansionary impact on output from a tax shock has become smaller and turned somewhat contractionary. This may likely reflect improvements in public administration and reforms undertaken following the Asian financial crisis (Nasution, 2003). On the other hand, the smaller contraction (improvement) in output response to a spending shock since 2000 may also reflect improvements in macroeconomic management and economic fundamentals of the country since the period.

In Malaysia, the expansionary response of output to a tax shock is most evident during the two major crisis periods. In normal times, however, the standard evidence of a contractionary impact on output largely prevails. Taken together, it is quite clear that the overall result from the Blanchard and Perotti SVAR is dominated by the crisis effects. More importantly, during crises when there is greater uncertainty and heightened risk premium, tax cuts may have been interpreted as weakening the fiscal position of the government with which agents respond with excessive belt tightening (see Hemming et al., 2002). In terms of spending shock, the output response is largely muted plausibly due to leakage through the high trade openness as mentioned above. Interestingly, the implementation of capital controls in 1998 to insulate the economy from external disturbances did not seem to have any notable impact on the government’s intended efforts to jump-start the economy. On the contrary, evidence since then suggests the opposite.

Similarly for the Philippines, the expansionary response of output to a tax shock is also most prevalent during the two crisis periods. Yet in the Philippines’s case, the effect is more pronounced – lasting more than a few years around the crisis periods. This seems to reflect agents’ greater concerns over the country’s perennial fiscal weakness and high public debt level vis-à-vis to, say, Malaysia’s. In addition, there is evidence from our VAR model to suggest that taxes are procyclical to output growth as the impact of a positive output shock leads to rising taxes. IMF (2009) also reinforces this, where it is shown that negative fiscal stance (deficit) often corresponds with a positive output gap. This means with the exception of a few years after the Asian financial crisis, where the output gaps turned negative, had higher taxes been implemented during the good times, output would have been boosted further as households and firms see these as real commitments to better manage the country’s fiscal balance and the high-debt level. In terms of the spending shock, the largely negative output response after 2000 (compared with the earlier decade) coincides with the period of higher public debt, especially after the Asian financial crisis and larger budget deficit. From 1994 to 1997, the country actually enjoyed a rare period of fiscal surplus, while the deficit in the early 1990s was smaller around 1–3% of GDP. Interestingly, the improvement in the output response in recent years (Fig. 2) coincided with the improvements in the public debt level and the budget deficit.

In Singapore, the expansionary response of output to a tax shock is most evident during the recent global financial crisis, which also happens to be the sharpest recession in its history. Given this extreme uncertain environment coupled with an export collapse, there was undoubtedly a heightened fear and anxiety which forced households to save more and firms to lay off workers to reduce spending and investment. Tax cuts implemented could reinforce this negative sentiment and pulled

¹⁸ Nasution also makes the point that Indonesia’s fiscal policy was directed to maintain a balanced budget. If this means when output growth rises too much, government spending is reduced and taxes are increased accordingly, then there is some evidence from the Cholesky VAR to support this.

output down. On the other hand, there is some evidence to show that during the two crisis periods, government spending was more effective in expanding output.

In Thailand, too, the expansionary response of output to a tax shock is most evident during the recent global financial crisis. However, in Thailand's case, it was the second most severe recession in history after the Asian financial crisis. Coupled with the long-drawn political crisis, the same loss of confidence and heightened fear as in Singapore would have prevailed in Thailand. This possibly explains the evidence of a larger output expansion during this period. On the other hand, there is some evidence that government spending during the Asian financial crisis was more effective especially at the 2-year horizon. This being the deepest recession in its history clearly suggests that there was ample capacity in the economy, which provided a bigger stimulus bang for fiscal policy.

6. Conclusion

In this paper, we investigate whether fiscal policy is an effective macroeconomic tool in influencing output in the five main ASEAN countries. Surprisingly, there has been very little empirical work done on these countries. Judging from the active use of discretionary fiscal stimulus especially during the recent global financial crisis, it would seem that policymakers have unequivocal confidence in its efficacy. What we find, however, is quite different. Our Blanchard and Perotti SVAR model shows for government spending, the overall impact on output is largely benign – the impact fiscal multiplier is very much less than one and statistically insignificant. This is true in all ASEAN5. In the case of tax measures, a consistent pattern of output expansion with fiscal contraction is observed, although the result is only statistically significant in Indonesia and Thailand.

The literature offers several explanations. Being small and highly open economies, Singapore, Malaysia, Thailand, and to a lesser extent the Philippines and Indonesia, are very susceptible to fiscal stimulus that leaks out through higher imports. Coupled with the adoption of a more flexible exchange rate regime especially after the Asian financial crisis, the leakage would have been greater.

More important, fiscal credibility characterized by a good track record of budget balances and low public debt level is key to policy effectiveness. The lack of fiscal credibility is a major factor that contributes to the expansionary fiscal contraction phenomenon observed in many other studies. Among the ASEAN5, many have run persistent budget deficits with the exception of Singapore. Their public debt levels may be considered as comfortable for developed countries, but not for developing countries epitomized by weak fiscal management and institutions, and a relatively small tax base. The Philippines's fiscal weaknesses are well known and including Indonesia are two countries that have faced sovereign debt problems and restructured their debt. Thailand came close during the Asian financial crisis and Malaysia has shown signs of weakness.

While the Blanchard and Perotti SVAR only summarizes the dominant influence over time, we suspect there may be times when either the government spending or tax change is more or less effective. To do this, we employ a time-varying VAR model and find that the expansionary impact of output to a positive tax shock in most countries is most prominent during crisis periods. In the Philippines, for example, this is most obvious, in both the Asian financial crisis and the global financial crisis lasting quite a few years. Similarly, evidence is found in Malaysia, but not as long lasting, and in Thailand and Singapore, mostly during the global financial crisis. In Indonesia, this phenomenon is most prevalent before the Asian financial crisis, but has improved since then in line with the fiscal reforms undertaken.

On the other hand, in terms of the impact on output of a government spending shock, Indonesia's fiscal policy effectiveness appears to have improved since the Asian financial crisis in line with improved macroeconomic management. The Philippines meanwhile follows the opposite path of deteriorating government spending impact on output. Given its weak fiscal credibility and further rise in public debt level and persistent budget deficit since the Asian financial crisis, these actions do not bode well with market confidence. In Malaysia, too, while the impact on output of a government spending shock is small and largely positive before the Asian financial crisis, it has turned somewhat negative since then. This may have to do with the introduction of capital controls during the Asian financial crisis and subsequent fiscal weaknesses. In Thailand, there is more evidence to show government spending can be countercyclical when it is tried during crisis periods.

In sum, we do find that fiscal spending can be effective at times in some countries, yet the evidence is by no means overwhelming and the multiplier is less than one. In contrast, tax cuts do not seem to have the same effect. On the contrary, tax hikes seem to boost output especially during crisis periods. Perhaps these actions are interpreted as greater fiscal responsibility and credible efforts that boost confidence which in turn spur private consumption and investment. Reductions in government spending may not have the same impact because government spending is often viewed as essential for development and future economic growth. Yet because of the differences in economic development and fiscal strengths, these reasons cannot satisfactorily explain the results for both Singapore and say the Philippines together. Admittedly, this is one area where future research can be done through a case study approach looking at specific tax measures in individual countries and examining their impact on consumption, investment and/or growth. In addition, for future research, there are at least two ways in which the VAR model can be improved. First, it would be useful to account for the role of monetary policy and the exchange rate, which is important for the transmission of external shocks to a small open economy. Second, a country's outstanding debt level may have important implications for the efficacy of fiscal policy. For example, [Caprioli and Momigliano \(2011\)](#) find that the

inclusion of such variable not only increases the estimation accuracy of the VAR model, but also produces a stronger impact of fiscal policy on economic activity.¹⁹

Appendix A. Data description and sources

Sources and descriptions of variables used are shown below. The population data are obtained from World Bank's *World Development Indicators*. World real GDP per capita (NGDPRPC) is obtained from the International Monetary Fund's September 2011 *World Economic Outlook* database.

Country	Period	GDP and Government Consumption	Taxes
Indonesia	Q1:1993 to Q4:2009	CEIC: Table ID.A01: Gross Domestic Product: By Expenditure: Current Price; Table ID.A02: Gross Domestic Product: By Expenditure: 2000 Price; Table ID.A04: Gross Domestic Product: By Expenditure: 1993 Price.	CEIC: Table ID.F05: Central Government Operations: Total Revenue (M1:2000–M12:2009). <i>Note:</i> Data prior to 2001 were sourced from Jha et al. (2010). Using a temporal disaggregation method, the authors converted the annual data to quarterly based on the quarterly government consumption available from national accounts.
Malaysia	Q1:1992 to Q3:2009	CEIC: Table MY.A01: 2000 Base: GDP By Expenditure: Current Price; Table MY.A03: 2000 Base: GDP by Expenditure: 2000 Price; Table MY.A13: 1987 Base: GDP By Expenditure: Current Price; Table MY.A15: 1987 Base: GDP by Expenditure: 1987 Price.	<i>Monthly Statistical Bulletin</i> , Bank Negara Malaysia: Table 6.2 Federal Government Revenue (http://www.bnm.gov.my/files/publication/msb/2009/11/xls/6.2.xls); Table 6.3 Federal Government Expenditure (http://www.bnm.gov.my/files/publication/msb/2009/11/xls/6.3.xls) <i>Note:</i> Data not available on the website are obtained separately from Bank Negara Malaysia.
Philippines	Q1:1990 to Q4:2009	CEIC: Table PH.A01: GDP by Expenditure: Current Price; Table PH.A02: GDP by Expenditure: 1985 Price.	CEIC: Table PH.F01: National Government Revenue, Expenditure and Financing.
Singapore	Q1:1990 to Q4:2009	CEIC: Table SG.A03: GDP by Expenditure: 2000 Price: Chain Linked; Table SG.A06: GDP by Expenditure: 1995 Price: Chain Linked; Table SG.A10: GDP by Industry: SSIC 2005: Current Price.	CEIC: Table SG.F04: Government Operating Revenue.
Thailand	Q1:1993 to Q4:2009	CEIC: Table TH.A01: Gross Domestic Product: By Expenditure: Current Price; Table TH.A03: Gross Domestic Product: By Expenditure: 1988 Price.	CEIC: Table TH.F01: Government Revenue: Ministry of Finance.

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¹⁹ We very much appreciate an anonymous referee for pointing this out.

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